

SURENDRA KUMAR REDDY RUKA

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Objective

Enthusiastic Embedded Systems Engineer with hands-on experience in STM32-based firmware, real-time communication systems, and hardware design. Skilled in C, C++, and embedded driver development for industrial and IoT applications. Adept at creating robust, efficient, and scalable embedded solutions while ensuring performance optimization and system reliability.

Skills

Programming :	C, C++, Embedded C, Python
Microcontrollers :	STM32 (F1/F3/F4/F7 series), Arduino, ESP32
Protocols :	UART, SPI, I2C, RS485, CAN, Modbus RTU/TCP, MQTT, Bluetooth, LoRa
Embedded Systems :	FreeRTOS, Device Drivers, Interrupts, Timers, ADC/DAC, Sensor Integration
PCB Tools :	KiCad, Altium Designer, DipTrace, Eagle
Dev Tools :	STM32CubeIDE, STM32CubeProgrammer, Arduino IDE, Git, GitHub
Other :	SCADA, DWIN GUI

Experience

Embedded System Engineer

Radiogeect Digital Pvt Ltd – Jamshedpur, India

April 2024 – Present

- Developed **Radiowave Remote IO**, a wireless P2P transceiver system using LoRa with a range of 10 km.
- Engineered bidirectional communication for analog and digital data transmission using **STM32F407**.
- Designed a 4-channel process display system utilizing **STM32F373** for simultaneous data visualization.
- Integrated **ADS1115** for analog input acquisition and **MCP4725** for 4–20 mA output transmission.
- Created an energy-saving system using **LD1115H radar** for efficient human presence monitoring.
- Designed a **UWB-based real-time position tracking system** for EOT cranes using STM32 and FreeRTOS.
- Built a **radar-based vehicle speed monitoring system** using LD2415H and ESP32 with real-time display on P4 LED matrix.

Embedded System Developer Intern

Radiogeect Digital Pvt Ltd – Jamshedpur, India

July 2023 – March 2024

- Acquired hands-on experience in embedded systems through a nine-month internship.
- Developed firmware for a process controller using **STM32F373C8T6**, interfacing **ADS1115 ADC** and **MAX485** for RS485.
- Designed custom PCBs for process controller with 4–20 mA / 1–5 V inputs and a 4-inch display.
- Created custom libraries for driving **LED matrix displays** (P10, P4, P2.5) with **STM32F407VGT6**.
- Designed RGB and single-color LED matrix PCBs using **STM32F103C8T6**.
- Built temperature/humidity display using **SHT20 sensors** and RS485 communication.
- Processed sensor data to generate 4–20 mA analog outputs.
- Developed anemometer display by interfacing wind speed sensor with STM32F373C8T6.

Education

B.Tech – Electronics & Communication Engineering

Rajiv Gandhi University of Knowledge Technologies, Srikakulam
GPA: 8.46

April 2024

Pre-University Course – MPC

Rajiv Gandhi University of Knowledge Technologies, Srikakulam
GPA: 8.0

May 2020

Class X – SSC Board

Zilla Parishad High School, R.V. Palli
GPA: 9.8

April 2018

Academic Projects

Amplifying Education System with Real-Time Updates – STM32, LoRa, GSM

Wireless classroom communication system enabling real-time announcements and timetable automation.

Accident Prevention System – Arduino, Eye-Blink Sensor

Driver drowsiness detection system with alert and safety trigger based on eye-blink monitoring.

Home Automation System – ESP32, Blynk

Smart home controller via Blynk and ESP32 for remote switching of appliances over Wi-Fi.

STM32F405 Eval Board Design – KiCad, 4-layer PCB

Designed STM32F405 board with UART, SPI, I2C, USB, and GPIO interfaces for embedded prototyping.

Accomplishments

- **State-Level Chess Player** and **3-time District First Place** winner.
 - **Udemy Certifications:**
 - **Mastering Microcontroller and Embedded Driver Development (MCU1)** – Bare-metal driver development in Embedded C for STM32 (GPIO, I2C, SPI, USART).
 - **Mastering Microcontroller: Timers, PWM, CAN, Low Power (MCU2)** – STM32 HAL-based driver programming with Timers, RTC, PWM, CAN, and low-power modes.
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Languages

Telugu : Native Language
English : Intermediate
Hindi : Intermediate